

Pace, Not Fatigue: The Step Length–Frequency Partition Across Four Years and a Half-Ironman

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Abstract

Running speed is the product of step frequency (f) and step length (L): $v = f \times L$. Any change in speed must be absorbed by one term or the other. The question is whether fatigue shortens it beyond what that partition predicts — the running literature reports 2–5% stride shortening. Four years of wearable data (2023–2026: 402 runs, 40,376 thirty-second windows, including one Ironman 70.3 run split) were used to test it. Speed increases are carried almost entirely by step length ($\beta_L = 0.88$, $\beta_f = 0.12$). Once pace and gradient are controlled the partition does not shift: setting aside a warm-up effect confined to the first 30 minutes, the 95% interval bounds step-length change between -0.37% and $+0.18\%$ over 50 minutes, excluding the literature benchmark. In the 70.3, stride shortened in the final quarter and the drop in pace accounted for all of it. Also, there were two secondary findings: running off the bike raised cadence by roughly 3% at matched speed, and cadence at 3.0 m/s rose 2.9 steps/min between 2024 and 2026.

1 The question

The main identity for this paper is the following:

$$v = f \times L \tag{1}$$

Any change in running speed must be absorbed by either the step frequency or the step length, or both. This is what defines the partition: how much of a change in speed each of the two terms carries. Step length falling when pace falls is simply arithmetic, so saying that step length decreases late in a run is not a finding — it is bound to happen. The real question is whether the partition between the two parameters shifts under fatigue, beyond what slowing alone accounts for.

Olaya-Cuartero et al. (2024) report that stride length shortens by 2–5% under fatigue, while step frequency findings are mixed. Because a stride is two steps and both quantities are expressed as proportional change, that benchmark applies directly to step length. The idea that frequency holds constant is therefore a hypothesis that must be tested, not an assumption.

Operationally, this predicts a negative coefficient on time-into-run when step length is regressed on it with speed and gradient controlled: shortening that the partition does not account for. A 2–5% reduction across a run corresponds to roughly -0.4 to -1.0% per ten minutes over a 50-minute window. The study set out to report whether this effect was detected in the single-subject data, no matter whether the outcome proved to be negative, positive, or even totally absent.

2 Data and Method

The data used in this study is a Strava bulk export of Garmin FIT files from a single subject. The export contained 1,177 activities, of which 448 were runs, 446 had an accompanying file,

and 441 had a moving time longer than ten minutes. Excluding phone-recorded GPX activities, which carry no cadence data, left 408 activities, all of which parsed without failure. Cleaning reduced this to 402 activities, producing 40,376 thirty-second windows from 1,208,740 samples. Although the export spans 2020 to 2026, 2022 consists entirely of GPX files, 2020 contains a single run below the ten-minute threshold, and 2021 contributes only three runs, retained as context rather than for comparison. Year-on-year comparison therefore spans 2023 to 2026.

Samples with missing or zero cadence were removed, as were samples with a cadence below 67 steps/min per leg, which was classed as walking. The threshold was chosen from the cadence histogram, which is clearly bimodal: a walking mode at 58 steps/min per leg and a running mode at 80–84, with the density trough between them at 70. Note that step frequency is reported throughout as both legs, or twice the recorded cadence. The first 60 seconds of each run were also dropped, as GPS and cadence readings settle over the opening seconds. A window is defined as 30 seconds of moving time, so stopped and walking time does not count towards it.

Gradient was treated as a covariate rather than a filter. Gradient noise ($SD \approx 2.0\%$) exceeds terrain variation ($SD \approx 1.8\%$), and that noise is speed-dependent: the slowest windows pass a flat filter 34.7% of the time against 47.7% for the fastest. Filtering on gradient would therefore truncate the low end of the speed range over which the relationship is fitted.

Step length was computed as a ratio of means rather than a mean of per-sample ratios, which preserves Equation 1 at window level. Standard errors were clustered by activity and per-run slopes aggregated by t-test, making the run the unit of evidence rather than the window.

Data was recorded on a Garmin Forerunner 235 until a change to a Forerunner 255 during 2023. An HRM-Pro Plus chest strap, which reports running dynamics, was worn for 53% of runs in 2023 and 73% in 2024, but almost none thereafter. Comparing the two sensors within year, where both appear, the strap reads 0.77% (1.27 steps/min) low; strap-recorded runs are therefore excluded from the cadence trend in Section 5.

Every other decision taken in this project is recorded in the decision log accompanying the repository.

3 Analysis A: the speed partition

Analysis A found that $\beta_L = 0.8781$ (SE 0.0062) and $\beta_f = 0.1219$, estimated across 37,315 windows from 370 runs. Speed increases are carried almost entirely by step length: a 1% increase in speed corresponds to a 0.88% longer step and only a 0.12% higher cadence. By identity these two coefficients sum to exactly 1, since both regressions share the same design matrix; this is therefore a check on the arithmetic rather than evidence about running. The same caution applies to the R^2 of 0.965 for $\log L$ against $\log v$. Because $\log L = \log v - \log f$ and $\log f$ varies very little, $\log$ step length tracks $\log$ speed almost perfectly by construction, so a high R^2 here is guaranteed and should not be read as explanatory power.

When the subject increases speed from 2.5 to 3.5 m/s, a 40% increase, step length rises by 34.4% while cadence rises by only 4.2%. In absolute terms, step length goes from 0.930 m to 1.250 m, while cadence increases only from 161.2 steps/min to 168.0 steps/min.

The left of Figure 1 shows how step length changes with running speed and the right shows the same for step frequency. With 37,315 windows a conventional scatter plot would be unreadable, so the data is binned into hexagons shaded by how many windows fall into each, on a logarithmic scale. The cadence cloud is much wider than the step length cloud, meaning speed determines cadence far less strongly than it determines step length.

A naive interaction test across 2023–2026 gave $F = 3.15$, $p = 0.025$, which taken at face value would suggest the partition differs by year. This result should not be taken at face value, because the years do not span the same speeds: in 2023 the 95th percentile speed was 3.96 m/s, against only 3.49 m/s in 2025. That matters because the relationship is slightly curved. Adding a quadratic term gives a $\log v^2$ coefficient of -0.042 ($p = 0.068$), meaning the slope flattens

Speed decomposition: 37,315 gradient-adjusted 30 s windows

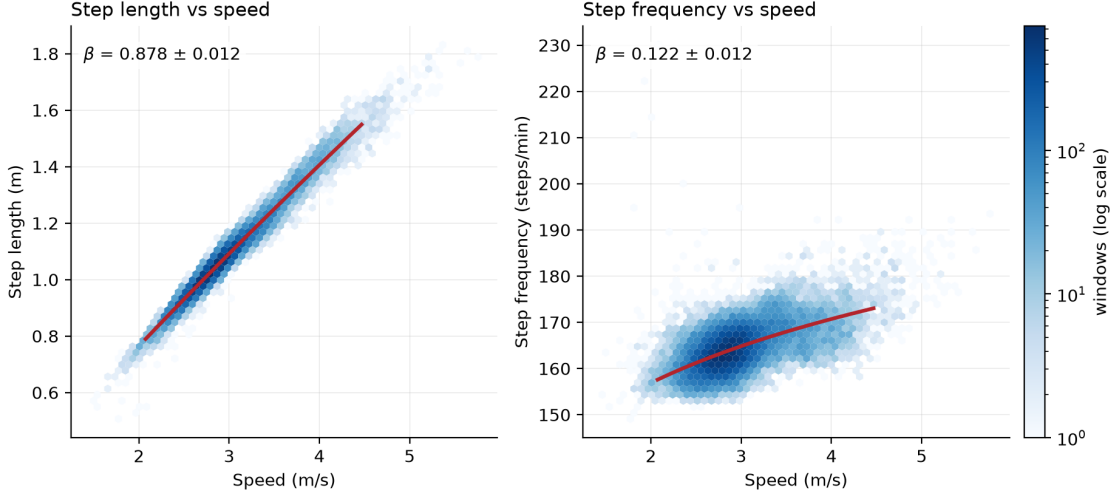


Figure 1: Step length (left) and step frequency (right) against running speed, across 37,315 gradient-adjusted 30-second windows. Windows are binned into hexagons shaded by count on a logarithmic scale; the red line is the fitted model evaluated at mean gradient. β is the elasticity of each quantity with respect to speed, and the two sum to 1 by the identity $v = f \times L$. A proportional speed change is carried 0.88 by step length and 0.12 by step frequency.

at higher speeds, so a year containing more fast running is fitted a lower slope regardless of whether anything about the runner changed. Restricting all years to the speed range they share (2.53–3.49 m/s) gives $F = 1.84$, $p = 0.139$, and β_L values of 0.898 in 2023, 0.886 in 2024, 0.895 in 2025 and 0.847 in 2026. No year effect is therefore reported. This restriction does come at a cost: matching the speed ranges cuts 2023 from 2,408 windows to 1,388, so part of the loss of significance is a loss of statistical power. The claim is that the year effect is not robust, rather than that it is absent.

Figure 2 shows both specifications side by side. Restricting the speed range widens the intervals, and the apparent separation between years in the unrestricted fit largely disappears.

As a further check, the flat-window subset originally planned for this analysis — windows with $|\text{gradient}| < 1\%$ — gives $\beta_L = 0.8785$ against the primary estimate of 0.8781. Treating gradient as a covariate rather than a filter therefore does not affect the result.

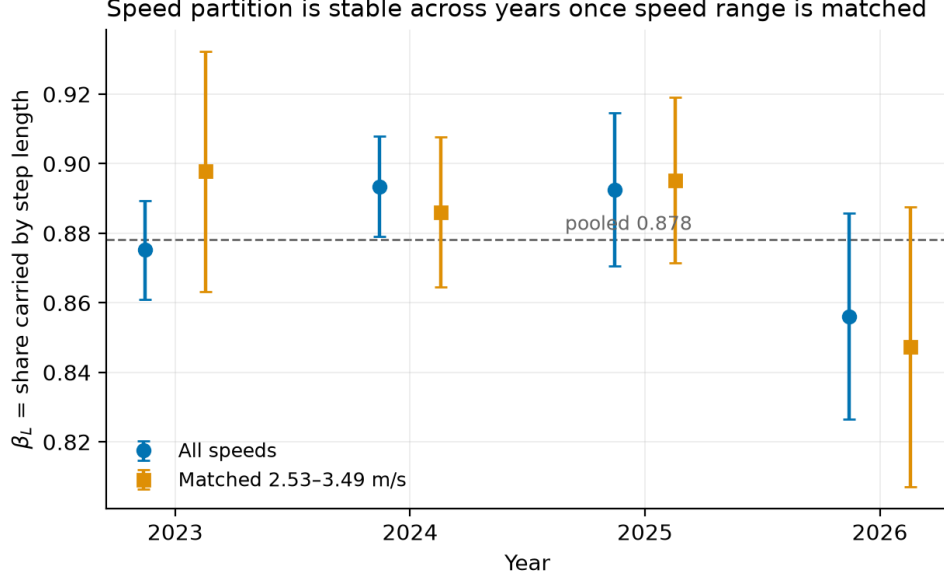


Figure 2: β_L by year under two specifications: all speeds (blue circles) and restricted to the 2.53–3.49 m/s range shared by all four years (orange squares). Error bars are 95% confidence intervals from standard errors clustered by activity. The dashed line marks the pooled estimate of 0.878. Matching the speed range removes most of the apparent separation between years.

4 Analysis B: fatigue within a run

4.1 The test

Analysis B was done by gathering all runs over 60 minutes, 93 in total, then filtering out 8 of them as they did not meet the gradient requirement. This left 85 runs and 16,198 windows. Each run was fitted separately and a one-sample t-test was then applied to the resulting 85 slopes. All 16,198 windows were not pooled into one model, as windows within the same run are highly correlated (same shoes, terrain, physiological state), so pooling would treat them as independent and give a confidence interval far too narrow. The run is the unit of evidence, not the window.

4.2 Controlling for pace

Table 1 reports the mean per-run slope under successive control specifications. The apparent effect in the uncontrolled fit collapses as soon as speed is included, and what remains is examined in Section 4.3. The left panel of Figure 3 shows these specifications graphically, excluding the first-30-minutes fit.

The analysis plan called for heart rate to be included alongside speed and gradient. This is the natural instinct, but it is not appropriate here. Heart rate rises through a long run largely because of the fatigue the analysis is trying to measure: cardiac drift is a symptom of the effect, not an independent influence on it. Including a mediator as a control absorbs part of the effect being estimated, so a genuine fatigue effect could be partly concealed by the very variable intended to clean the estimate up. Heart rate is therefore reported as a robustness check rather than as part of the primary specification. In this case the choice does not alter the conclusion: after the warm-up period the estimate is -0.018% per 10 minutes without heart rate and -0.040% with it, and neither is distinguishable from zero. Both are reported so that the specification chosen is visible rather than assumed.

Specification	Slope (%/10 min)	95% CI
<i>Full run</i>		
No controls	+0.780	[+0.451, +1.109]
+ speed	+0.093	[+0.051, +0.135]
+ speed, gradient	+0.098	[+0.055, +0.141]
+ speed, gradient, HR	+0.068	[+0.024, +0.111]
First 30 minutes only	+0.549	[+0.441, +0.657]
<i>After the first 30 minutes</i>		
No controls	+0.658	[+0.036, +1.281]
+ speed, gradient	−0.018	[−0.073, +0.036]
+ speed, gradient, HR	−0.040	[−0.095, +0.015]

Table 1: Mean change in step length, in percent per 10 minutes, under successive control specifications, across 85 runs of at least 60 minutes. Each run was fitted separately and the table reports the mean of the resulting 85 slopes; intervals are 95% confidence intervals across runs. Heart rate is included as a robustness check rather than a control, for the reason given in the text.

4.3 The warm-up

It was found that step length rises by around 0.55% per 10 minutes over the first 30 minutes at constant speed, and then goes flat. Two independent checks corroborate this. The first is duration: the effect dilutes as runs get longer, with 60–90 minute runs giving +0.126% per 10 minutes, 90–120 minute runs +0.095%, and runs over 120 minutes +0.007% ($p = 0.82$). This is exactly the pattern expected from a fixed early-run effect averaged over a longer window. The second is intensity: splitting runs into heart-rate tertiles gives +0.129, +0.076 and +0.088% per 10 minutes, showing no trend across tertiles, so the effect is not effort driven.

4.4 Bounding the effect

Excluding the first 30 minutes, across the remaining 50 minutes of a typical long run the interval bounds the total change in step length between −0.37% and +0.18%. The literature benchmark is a shortening of 2–5%, which this interval excludes. Only 5% of the 85 runs fall within the benchmark band, shown as the distribution in the right panel of Figure 3.

4.5 Testing the result

The cleaning of the data removes any walking. The issue with this is that if fatigue expresses itself as walk breaks rather than shortened running strides, those windows get deleted. This leaves a biased sample for later in runs, when the subject is fatigued. Table 2 illustrates that in the final third of a long run there is 2.5 times as much non-running time per second of running as in the first third, and that the running which does survive has longer strides.

Table 3 shows the slope, as a percentage per 10 minutes, for subsets defined by how often the subject was not running. The slope stays approximately the same in every subset; the only thing that changes is the confidence interval, which widens as the number of runs falls. Removing the suspected cause therefore does not remove the effect, so selection is not what produces it.

The evidence points against the selection explanation more directly still. The per-run slope correlates negatively with pause rate (−0.24). Selection would predict the opposite: deleting the poorer running late in a run leaves a better sample behind, so runs with more walking should show more inflated slopes, not less. A plausible reading is that runs with more pauses were

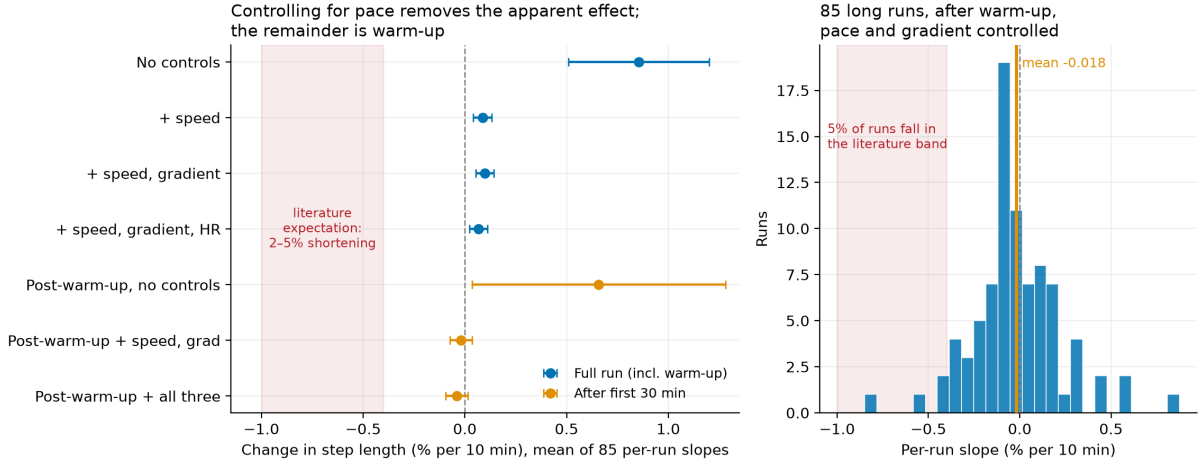


Figure 3: Left: mean change in step length per 10 minutes under successive control specifications, with 95% confidence intervals across the 85 runs. Blue points fit the whole run; orange points fit only the portion after the first 30 minutes. The shaded band marks the literature expectation of 2–5% shortening across a run, which corresponds to -0.4 to -1.0% per 10 minutes over a 50-minute window. Controlling for pace collapses the effect, and what survives is confined to the warm-up. Right: the distribution of the 85 individual post-warm-up slopes with pace and gradient controlled, centred on -0.018% per 10 minutes. Five percent of runs fall inside the literature band.

	First third	Middle third	Final third
Non-running seconds per running second	0.052	0.084	0.130
Windows following a gap	0.9%	1.7%	2.4%
Mean step length (m)	1.042	1.072	1.087

Table 2: Running continuity and step length across thirds of a long run, averaged over the 85 runs of at least 60 minutes. Non-running time accumulates 2.5 times faster in the final third, and the running that survives cleaning has progressively longer strides — the pattern a selection effect would produce.

Subset	Runs	Slope (%/10 min)	95% CI
All long runs	85	+0.098	[+0.055, +0.141]
Pause rate < 0.05	44	+0.113	[+0.051, +0.175]
Pause rate < 0.02	26	+0.104	[+0.008, +0.199]
Pause rate < 0.02, no gaps	18	+0.110	[−0.023, +0.242]

Table 3: Change in step length per 10 minutes, with pace and gradient controlled, for progressively more continuous subsets of the long runs. Pause rate is non-running time per second of running. The point estimate is stable across all four subsets; only the interval widens as the number of runs falls.

runs in which the subject was more challenged, and so was running in a more fatigued state throughout.

Retention is defined as the fraction of a run’s moving time that survived the cleaning pipeline. Restricting to runs with retention above 0.70 or above 0.80, and separately excluding runs recorded with the chest strap, gives post-warm-up slopes of -0.020 , -0.021 and -0.038 per 10 minutes respectively. Each of these confidence intervals includes zero, so the null result holds under all three restrictions.

4.6 The 70.3 run split

The subject competed in Ironman 70.3 Belgrade on the 13th of September 2026. The run split was a recorded 20.89 km (actual race distance was 21.1 km) in 107 minutes of moving time. This gave 211 windows with a mean speed of 3.26 m/s and a heart rate range of 148–171 bpm, averaging 157 bpm.

Table 4 splits the run into quarters and shows that the pace holds through to 15 km and then fades over the final quarter, as seen in the upper panel of Figure 4. Step length decreases by 3.4% over that final quarter.

Quarter	End (km)	Speed (m/s)	Step length (m)	Cadence (spm)	HR
1	5.0	3.275	1.125	174.7	153
2	10.1	3.275	1.135	173.1	157
3	15.1	3.310	1.148	173.0	160
4	19.9	3.169	1.109	171.5	159

Table 4: The 70.3 run split by quarter. Pace is held through three quarters and falls by 4.3% in the fourth, with step length falling 3.4% alongside it.

The naive slope after 30 minutes is -0.569% per 10 minutes, which matches the fatigue trends described in the literature. This estimate, however, is uncontrolled. Once speed and gradient are controlled the slope becomes $+0.094\%$ per 10 minutes. The shortening in step length is entirely accounted for by the drop in pace, meaning the partition did not shift.

The controlled slope puts the race in the 73rd percentile of the 85 training-run slopes. The residual-against-distance slope after the first 5 km is $+0.055\%$ per km, totalling $+0.83\%$ over the remaining 15 km; this is visible as the near-flat trace in the lower panel of Figure 4. The value is very small and the wrong sign for fatigue, which would produce a negative slope.

This answers the initial question of the report in a case where there is definite fatigue: step length fell exactly as much as slowing predicted, and no more.

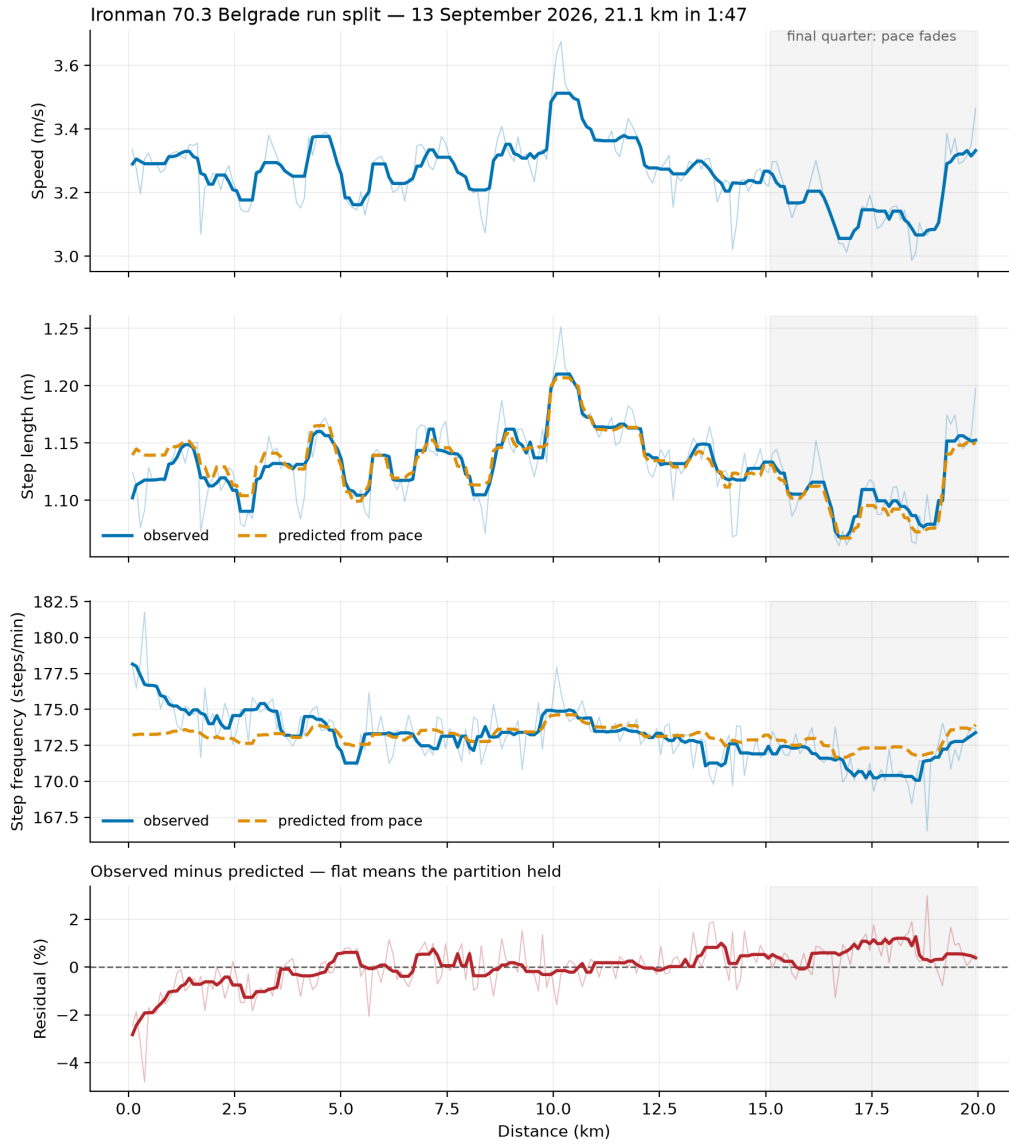


Figure 4: Ironman 70.3 Belgrade run split, 13 September 2026: 21.1 km in 1:47. Panels show speed, step length and step frequency against distance; faint lines are individual 30-second windows and bold lines a 7-window rolling median. The dashed orange line is the Analysis A prediction from pace and gradient, corrected to the run's own mean. The bottom panel shows observed minus predicted step length, where a flat trace indicates the partition held. The shaded region marks the final quarter, over which pace fades.

5 What did move the partition

Analysis B found that nothing moved the partition within a run, but two activities sit outside this pattern: the run legs of two triathlons. At matched speeds, cadence was elevated by +3.09% in an Olympic-distance run leg (14 July 2024) and by +3.33% in the Ironman 70.3, relative to matched comparison runs. This shift cannot be attributed to fatigue, because the difference is present from the first kilometre and does not drift. The Olympic-distance leg was strap-matched to 89 comparison runs from 2024, so the sensor difference cancels out. The strap was used on only two other runs in 2026, which is not enough to match the 70.3, and since the strap reads low the wrist-equivalent gap is closer to +4%.

Both of these runs were in race conditions, where arousal, race shoes and pacing are all confounded with running off the bike. There are no brick sessions in the dataset, so on two events the effect of the bike cannot be separated from the effect of racing. What the two do show is that the method detects a partition shift when one is present.

The cadence trend over the years was also plotted as a secondary finding. Figure 5 shows cadence at 3.0 m/s for each year in the upper panel, with total running volume in the lower bar chart: 279, 1,152, 1,450 and 857 km, the last covering January to September only. There is a rise of +2.88 spm from 2024 to 2026, with the jump concentrated in 2026. Without correction the rise was initially found to be +4.25 spm. Around 30% of the apparent trend was a change in the chest-strap sensor rather than the subject, which was spotted and accounted for.

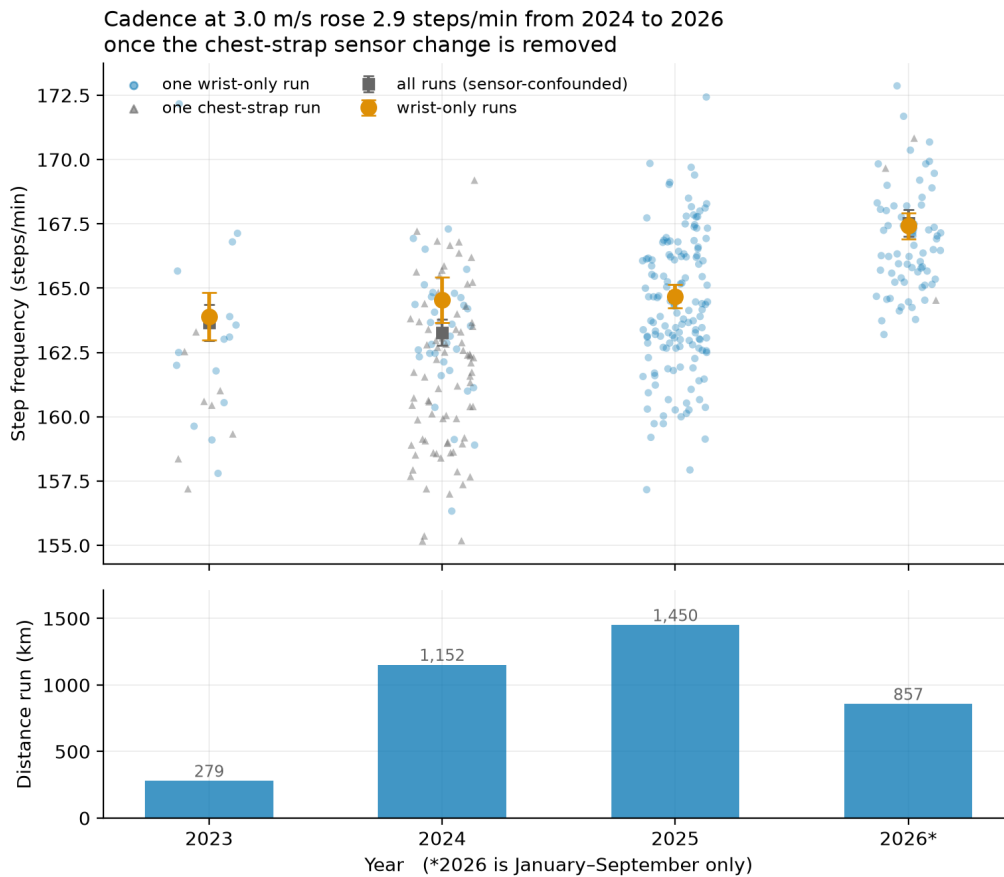


Figure 5: Cadence at 3.0 m/s by year. The upper panel shows the wrist-only series alongside the uncorrected series that pools wrist and chest-strap runs; the gap between them is the sensor artefact, which accounts for roughly a third of the apparent rise. The lower panel shows annual running volume, with 2026 covering January to September only.

6 Limitations

The most obvious limitation of this study is that there is only one subject, the author. There is no force plate and no motion capture, so there is no ground truth for any derived quantity. The findings describe a single runner’s gait and cannot be generalised to other runners.

The measurement chain is unvalidated. Step frequency comes from an accelerometer at the wrist or chest, and speed comes from GPS. Step length is derived from these two, as speed divided by step frequency. Garmin reports its own step length field, and the derived value agrees with it to -0.25% . That agreement confirms the arithmetic and nothing more, because Garmin almost certainly derives its field from the same two sensors: two calculations from the same inputs will agree whether or not either sensor is accurate.

The walking filter is the main constraint on the headline result. Late in a long run the subject accumulates roughly 2.5 times more non-running time per second of running, and those samples are removed by design. The null from Analysis B is therefore specifically that step length does not shorten beyond the partition *while running*. The design is structurally incapable of seeing fatigue that is expressed as stopping to walk, which is a plausible fatigue response in its own right.

Running off the bike cannot be separated from racing. Both observations are race efforts, so arousal, race shoes and pacing strategy are confounded with the bike leg. There are no brick sessions anywhere in the dataset, and there are two events in total.

Over the four years of data collection the subject changed shoes, body mass, training volume and training style, none of which is recorded. This is why the cadence trend is the weakest finding reported here: the sensor artefact could be identified and corrected, but these cannot.

Nothing in this data measures joint angles, ground reaction forces or tissue loading. No claim about kinematics, loading or injury risk follows from any of it.

References

- Olaya-Cuartero, J., et al. (2024). Influence of Fatigue on the Modification of Biomechanical Parameters in Endurance Running: A Systematic Review. *International Journal of Exercise Science*, 17(1), 1377–1391.
- Zandbergen, M.A., et al. (2023). Effects of level running-induced fatigue on running kinematics: A systematic review and meta-analysis. *Gait Posture*, 99, 60–75.